Example 8.4

The following are the information about the number of persons who are affected by Diabetes and Lung Cancer and the number of persons died due to each cause of death during a calendar year in two different districts:

Cause of Death	District A		District B	
	No. of Persons		No. of Persons	
	Affected	Died	Affected	Died
Diabetes	20,000	325	22,000	400
Lung Cancer	19500	300	21,225	380

Find the Illness specific death rates for the two districts. Also, compare health conditions of both the districts with reference to these two causes of death. Assume that a person affected by Diabetes is not affected by Lung Cancer and *vice-versa*.

Solution:

The *SDR* due to the two causes of death are calculated as follows:

SDR of District A

$$SDR_{Diabetes} = \frac{D_{Diabetes}}{P_{Diabetes}} \times 1000$$

$$= \frac{325}{20000} \times 1000$$

$$SDR_{Diabetes} = 16.25 \text{ per thousand}$$

$$SDR_{Lung\ Cancer} = \frac{D_{Lung\ Cancer}}{P_{Lung\ Cancer}} \times 1000$$

$$= \frac{300}{19500} \times 1000$$

$$SDR_{Lung\ Cancer} = 15.38 \text{ per thousand}$$

SDR of District B

$$SDR_{Diabetes} = \frac{400}{22000} \times 1000$$

$$SDR_{Diabetes} = 18.18 \text{ per thousand}$$

$$SDR_{Lung\ Cancer} = \frac{380}{21225} \times 1000$$

$$SDR_{Lung\ Cnacer} = 17.90 \text{ per thousand}$$

In both the districts, death rates are more due to Diabetes in comparison with Lung Cancer. Among the two districts, District B has relatively more death rate due to both Diabetes and Lung Cancer.

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